INDEPENDENT CLOUD-READINESS REVIEW

PreFlight: evidence before the estimate

An offline review of a Guidewire-style policy estate. It inventories what exists, links every claim to file evidence, and names every unknown — before a migration estimate is signed.

- **Claim → evidence**
- **Unknown → named question**
- **Change → pending human**

Reality boundary: a synthetic rehearsal, not an actual migration. No tenant. No compile. Not official Cloud Assurance.

10 minutes · 12 slides · run R-DEMO-001 · 300 synthetic files · offline

Review packet

PKT-R-DEMO-001

SYNTHETIC

Files hashed	300
Evidence records	101
Links on PA-VEH-01	31
Findings / dispositions	24 / 12
Human gates open	11

OFFICIAL CLOUD ASSURANCE: NOT RUN

Local draft. Not a quote, certification, or deployment. Every change is pending.

THE PROBLEM

ESTIMATE-MOVING SCOPE

The estimate is the promise. The inventory is the proof.

Scope hides under a migration number: files never classified, features never traced, and unknowns nobody owns. When the inventory moves, the estimate moves with it.

Estimate ≠ **Inventory**

a number with a date — the files, links, and gaps behind it

01 What exists?
Files, hashes, parser status, and exclusions. Nothing quietly dropped.

02 Which feature does it serve?
PCF → entity → typelist → template / Gosu → tests, linked across artefacts.

03 What is still unknown?
An owner, an impact range, and the one question that would close the gap.

THE THREE PROMISES

01 Every claim has evidence.
Path, locator, excerpt hash, method, and a claim-strength label. If it cannot be pointed to, it is not a finding.

02 Every unknown has a name.
An assumption with an owner, an impact range, and a question that can be answered or rejected.

03 Every change waits for a human.
Drafts start pending. No agent approves a change, resolves an assumption, or deploys anything.

What changes: scope becomes countable · dispositions become explicit · unknowns become questions. The estimate gets a defensible inventory underneath it.

R-DEMO-001

Five agents. One evidence chain. No authority to deploy.

The feature path is a straight handoff chain, not a loop. Each agent receives the prior artifact, adds a check or an artifact, and leaves a record. The run ends with human gates open — not approval.

01

Scout

INVENTORY & EVIDENCE

300 files **101** evidence **31** links

24 findings · 12 dispositions

›

02

Architect

DRAFT WORK PACKAGES

5 packages **5** targets **0** approved

schema · mapping · updater · codegen · API

›

03

Guardian

OPEN HUMAN GATES

11 gates **3 + 8** **6** high

3 assumptions + 8 rule groups · 0 resolved

›

04

Critic

INDEPENDENT CHECKS

7/7 checks **PASS** **0** auto-approved

Existence · parse · uniqueness · links · approval

›

05

Scribe

SEAL THE BUNDLE

8 sealed files **12** events **1** manifest

completed_with_gates · sealed ≠ approved

Chain, not loop.
The feature path ends in Scribe. The codebase lab is the cyclic variant, with Scout → Architect ↔ Guardian / Critic → Planner.

Scribe vs Planner.
Scribe seals the feature packet and log. Planner assembles the codebase patch plan and SARIF. Neither compiles or deploys.

Status: completed_with_gates · 11 open human gates · 0 resolved · 0 auto-approved · 0 deployed. “Green” is a bounded agent verdict, never deployment approval.

5 PACKAGES · 11 GATES · 7/7

Each agent received evidence. Each left a review artifact — not an approval.

Scout turns a snapshot into evidence. Architect turns dispositions into five local files. Guardian opens gates, Critic checks package integrity, Scribe seals the bundle. Every change remains pending.

AGENT	RECEIVED	PRODUCED	WHY IT IS STILL PENDING
Scout	300-file snapshot, rule pack	**101** evidence · **31** links · **24** findings · **12** dispositions	Findings are candidates, not confirmed violations.
Architect	12 dispositions with evidence IDs	**5** packages · **5** target files · all `draft_ready`	All approvals `pending`; 0 `_gen` writes.
Guardian	Packages, 3 assumptions, 8 rule groups	**11** gates · **6** high · **0** resolved	A gate is a named human decision, not a score.
Critic	5 packages + evidence links	**7/7** integrity checks · verdict PASS	Checks existence, parse, uniqueness — not correctness.
Scribe	Verified bundle + 12-event log	**8** sealed files · manifest · `completed_with_gates`	Sealed means tamper-evident, not approved.

Scribe
Feature path. Seals 5 targets + plan + report + log. 8 listed files. No compile.

Planner
Codebase path. Assembles patches + plan + SARIF. No compile; gates stay open.

Why every change is pending. Approval is a human action. The agents can draft, gate, check, and seal. None can approve, resolve an assumption, or deploy.

TWO DIFFERENT 11S

11 human gates in the migrator chain (3 assumptions + 8 rule groups). **11 audit events** in the post-decision review packet. The seed packet has 9 events and 3 assumptions; after two decisions it has 11 events and 4 assumptions. They are not interchangeable.

`8 listed bundle files = 5 target drafts + plan + report + 12-event log. The manifest is the seal wrapper.`

SYNTHETIC QUSAR 2026.08 SNAPSHOTS

Two estates, same crew, different blocker mix.

Cascade is monoline Personal Auto with a narrow integration surface. Harbor is multi-line P&C with claims, billing, and coastal underwriting crossing application boundaries. Both end machine-green with human gates open.

MONOLINE PERSONAL AUTO · SYNTHETIC

Cascade Mutual

GREEN_WITH_HUMAN_GATES

18	22	2	10
FINDINGS	PATCHES	ITERATIONS	HUMAN GATES
10 error · 8 warning	all pending	error-first, then full	all open

Structural score 100/100; human-gate-adjusted 74/100. The gap is the point. Every patch is a reviewable diff, not a deployment.

MULTI-LINE REGIONAL P&C · SYNTHETIC

Harbor P&C

GREEN_WITH_HUMAN_GATES

20	21	2	11
FINDINGS	PATCHES	ITERATIONS	HUMAN GATES
11 error · 9 warning	all pending	error-first, then full	all open

Structural score 100/100; human-gate-adjusted 71/100. APD deliverable is type-level naming preparation only — not full APD conversion or field mapping.



App outcome strips for the same synthetic runs: Cascade 18/22/2/10 · Harbor 20/21/2/11. The app says “blockers found” and “patches drafted”; this deck says findings and patches.

16 rule families · SARIF 2.1.0 advisory · 13 rules fired on Cascade, 15 on Harbor · **machine green ≠ deployment approval** · 0 compiled · 0 deployed.

CODE-LEVEL REFACTOR · DRAFTS, NOT DEPLOYABLES

DRAFTS ARE NOT COMPILED

The lab shows the target shape. It does not compile it.

One source pattern becomes a candidate Gosu Query service. A legacy JMS plugin or servlet becomes an App Events publication or Cloud API route draft. Sixteen rule families drive the scanner; SARIF 2.1.0 carries advisory results.

SQL → Gosu Query

BEFORE · DIRECT JDBC + CONCATENATED SQL

```
var conn = DriverManager.getConnection(
  "jdbc:cascade:rating")
var stmt = conn.createStatement()
stmt.executeQuery(
  "SELECT factor FROM cascade_rating " +
  "WHERE vin = '" + vin + "'")
```

DRAFT · TYPED GOSU QUERY / APPLICATION API

```
class VehicleRatingQueryService {
  function load() : List<VehicleRating> {
    return Query.make(VehicleRating)
      .select().toList()
  }
}
```

Corrected class name: `VehicleRatingQueryService`. The app generator currently appends `QueryService` again, producing `VehicleRatingQueryQueryService`; the doubled name is a generator artifact. Semantic equivalence is a human gate; the draft is not compiled.

JMS + servlet → App Events / Cloud API

BEFORE · LEGACY TRANSPORT + CUSTOM SERVLET

```
class PolicyMessagingPlugin
  implements IMessageTransport { … }
class DealerFeedServlet
  extends HttpServlet { … }
```

DRAFT · EVENT PUBLICATION + API ROUTE

```
app_events:
  name: policy-messaging-publication
  delivery: at-least-once
cloud_api:
  base_path: /ext/v1/dealer-feed
  auth: service-user
```

Not a drop-in. App Events is asynchronous and at-least-once; ordering, consumer acknowledgement, route auth, and consumer cut-over remain human gates. The generated configuration is a local draft, not deployable.

16 rule families · **SARIF 2.1.0** advisory: `official_cloud_assurance: false` · `synthetic: true` · 18 Cascade / 20 Harbor results · every patch `pending` .

SIMULATED, NOT A PUBLISHED RELEASE

A green signal is release-bound. Revalidate, roll back, or defer.

An assessment can outlive the release it was written against. PreFlight bumps the assessment generation, invalidates the stale green signal, and records one of three resume choices. A human still decides.

DETECTION · GENERATION 1 → 2

GENERATION 1		SCENARIO		GENERATION 2
green	→	APD 24→20	→	~~green invalidated~~
release-bound		2 patches each		revalidation_required

Simulated scenario: `GWCP-APD-NAMING-2027.04` moves the APD type-name limit from 24 to 20 characters. The 2027.04 release and the 20-character limit are PreFlight simulations, not published Guidewire release notes.

APD NAMING · 2 / 2 | APP EVENTS V2 · 3 / 3 | GATEWAY MTLS · 7 / 6 | REMOVED GOSU API · 14 / 14 | CLOUD API V2 · 4 / 3

01 · detect
A simulated release change arrives with a new rule pack.

02 · invalidate
The prior green becomes stale; affected patches are matched.

03 · decide
A human chooses revalidate, rollback, or defer.

RESUME CHOICES · ALL CLEAR THE GREEN

Revalidate	Re-run Architect with the new rule pack. Records `resume_requested`.
Rollback	Return to the last green checkpoint; keep the sealed bundle. Records `rolled_back_to_checkpoint`.
Defer	Pin the assessment release and record the risk owner. Records `upgrade_deferred`.

Non-cumulative exposure:

Cascade has 30 patch-exposures across 22 patches; Harbor has 28 across 21. One patch can appear in more than one scenario — never sum the rows to contradict the estate totals. Counts are exposure, not effort or defect counts.

TAMPER-EVIDENT · LOCAL

A claim is only as strong as its locator, its hash, and its open question.

PreFlight does not store “a finding.” It stores a claim with a source pointer, an excerpt hash, a method, a confidence label, and — where evidence runs out — an assumption with an owner. The packet and audit live in two separate chains.

EVIDENCE CARD · EV-0001

PATH	`config/web/pcf/policy/PAVehicleDV.pcf`
LOCATOR	`xpath /PCF/Screen/TypeKeyInput[1]:4`
METHOD	xml_xpath · confidence **parsed**
CLAIM	VERIFIED-DOC · caps what may be inferred
LINKS	**101** evidence records · **31** on PA-VEH-01

`excerpt_sha256 · d76bf9908b1f…`

The excerpt is re-read from the snapshot and hashed. A fabricated or drifting excerpt fails the locator-support audit. 101/101 links passed.

WHAT THE RECORD DOES NOT PROVE

- That the excerpt is semantically sufficient.
- That the decision is correct — only that it is inspectable.
- That official Cloud Assurance ran. It did not.

PACKET + AUDIT TRACEABILITY

`event 1` → `hash 1` → `event 2` → `…` → `head`

Packet hash binds the rendered packet; content digests bind findings, mappings, and evidence. Audit head: `read live`. Tamper-evident, not non-repudiation.

Review chain 9 seed → 11 post-decision **Migrator chain** 12 events

VERIFY LOCALLY

```
$ preflight verify-audit R-DEMO-001
OK (9 events, head …)
$ preflight verify-packet R-DEMO-001
OK - packet hash, content digests, audit chain verified
$ preflight verify-migration R-DEMO-001
OK - 12 events verified
```

Traceability: 5 packages → 11 human gates → 7/7 machine checks → 8 sealed files. **0 auto-approved.** Every change waits for a reviewer.

CLEAN DEMO VISUALIZATION · FULL-BLEED MAP

Open full visualization ↗

One evidence chain, two migration shapes, zero deployment.

FEATURE CHAIN · R-DEMO-001 · PA-VEH-01

Scout
300 files
101 evidence
31 links

Architect
5 packages
5 targets
all pending

Guardian
11 gates
3 + 8
6 high

Critic
7/7 checks
PASS
0 auto-approved

Scribe
8 sealed files
12-event log
sealed ≠ approved

CODEBASE LAB · SAME FIVE ROLES, CYCLIC REWORK

Scout
Cascade **18** findings
Harbor **20** findings
synthetic snapshots

Architect
22 / **21** patches
2 passes each
all pending

Guardian
13 machine checks
deterministic
gates open

Critic
7 checks
10/11 human gates open
no auto-approval

Planner
report + **SARIF 2.1.0**
review bundle
no compile

↺ rework to Architect if a machine check fails

AGENTS CAN
read · draft · gate · check · seal
No approval authority

→

A HUMAN DECIDES
approve · resolve · deploy
0 auto-approved · 0 deployed

11 feature gates **6** high **10/11** codebase gates **7/7** checks **8** sealed files **0** deployed **Official Cloud Assurance: not run**

Subtle pulse on agent dots; connector shimmer. Motion is disabled under `prefers-reduced-motion`. Counts are synthetic and run-specific.

ENGINEERING RIGOR

OFFLINE · NO TELEMETRY

The demo is the surface. The checks are the work.

The prototype is real code with tests, freeze hashes, audits, and a security contract. It stays offline: no external calls, no telemetry, no live tenant. Live reasoning is gated to synthetic fixtures.

current pytest suite
520 PASSED AT THIS SNAPSHOT · 2026-11-11

18/18
RECORDED FREEZE HASHES RECOMPUTE

0
EXTERNAL CALLS · TELEMETRY · LIVE TENANT

8
ADVERSARIAL FLAWS FOUND, FIXED, REGRESSION-TESTED

AUDITS AND CHECKS

- ✓ Independent layman + visual audit, round 1. This compact cut addresses the mobile-header, print-contrast, stale-evidence, and table-density findings.
- ✓ Internal adversarial review: 8 flaws fixed, including concurrent mapping writes, secret leakage, YAML alias bombs, and forged evidence scoring.
- ✓ Playwright golden path; 18/18 frozen hashes; 101/101 evidence links supported.
- ✓ Security contract: archive, XML, YAML, CSRF, CSP, redaction, and mobile overflow tests.

CONTROLS KEPT VISIBLE

127.0.0.1 BIND · HOST / ORIGIN ALLOWLIST · CSRF TOKEN · CSP SELF-ONLY · ZIP-SLIP / BOMB REJECTION · XML ENTITIES OFF · YAML TAGS REJECTED · HASH-CHAINED AUDIT · NO TELEMETRY · SYNTHETIC-ONLY LIVE GATE

Not a compiler, not a tenant connector, not official Cloud Assurance. The green signal is an internal structural verdict with human gates open.

Offline by construction: no external fonts, scripts, CDNs, or network calls in this deck · read-only snapshot · no customer data · no credentials.

REALITY BOUNDARY & PROOF LADDER

CURRENT STAGE: 1 OF 5

Real software. Synthetic estate. Unproven migration.

The scanner, evidence links, unified diffs, SARIF exporter, hash chain, and tests are genuinely implemented. The estate, insurer names, code, release scenarios, and evaluation slice are synthetic. Nothing was compiled, deployed, or officially assured.

WHAT IS REAL

- Deterministic scanner, evidence graph, unified diffs
- SARIF 2.1.0 advisory export and hash-chained audit
- Two synthetic migration archetypes and the test suite

WHAT IS NOT

- No real insurer snapshot, tenant, data model, or production data
- No Gosu/Java compile, runtime, or integration test
- No official Cloud Assurance; no deployment or cutover

WHAT WOULD MAKE IT A REAL MIGRATION

REAL READ-ONLY SNAPSHOT | COMPILE / TEST | OFFICIAL CLOUD ASSURANCE | SME SIGN-OFF | PRODUCTION CUTOVER + ROLLBACK

OFFICIAL CLOUD ASSURANCE: NOT RUN

01	**Synthetic rehearsal** Method is repeatable and evidence-linked. Current stage.	NOW
02	**Real read-only snapshot** Authorised estate, chain of custody, blind SME sample.	NEXT
03	**Compile / test in Guidewire dev** Target release, compiler logs, GUnit, smoke tests.	NOT DONE
04	**Official Cloud Assurance + SME sign-off** Official report, threshold, violation triage, named owners.	NOT DONE
05	**Production cutover + rollback** Deployment, reconciliation, rollback test, hypercare, outcomes.	NOT DONE

PCF
Page configuration XML.

GOSU
Guidewire's JVM language.

TYPELIST
Reusable code list.

ENTITY
Data-model object.

APD
Advanced Product Designer.

JMS / JDBC
Legacy messaging / direct SQL.

SARIF
Static-analysis results format.

LOB
Line of business.

CLOSE · THE STANDARD AND THE ASK

Every claim has evidence.
Every unknown has a name.
Every change waits for a human.

That is the standard PreFlight is trying to make ordinary: an estimate you can inspect, challenge, and sign — with the unknowns written down instead of discovered at week five.

Pilot estate
One authorised, read-only snapshot of one LOB. The next experiment is a real inventory, not another synthetic fixture.

Blind SME review
Two domain experts annotate a sample independently; disagreements are adjudicated. That is the human IAA the current evaluation lacks.

Hard feedback
Attack the claims, the packet, and the disposition model. The fastest way to improve this is to try to break it.

Pilot ask: one authorised read-only estate · two blind SME annotators · hard feedback. No customer data needed for the first conversation.

Saundarya Singh · independent research prototype **Repository** github.com/alikhan37544/Pre-Flight_Autonomous_Migration **License** MIT **Deck** docs/presentation/compact-10min.html

Independent academic research prototype. Not affiliated with, endorsed by, or certified by Guidewire Software, Inc. All fixtures are synthetic. Findings are pattern-based heuristics, not official Cloud Standards checks or legal/compliance advice.
Official Cloud Assurance: not run.